

4. Alkaloids

Definition

- A precise definition of the term 'alkaloid' (alkali-like) is somewhat difficult
- Typical alkaloids are derived from plant sources, they are basic,
- They contain one or more nitrogen atoms (usually in a heterocyclic ring)
- They show striking pharmacological effect on vertebrate animals
- More than 15,000 nitrogen containing secondary metabolites found in 20% of plants
- Most alkaloid are alkaline, at pH value commonly (7.2)

- **True alkaloids** are derived from amino acids and they share a heterocyclic ring with nitrogen and are highly reactive with biological system even at low doses
- **Protoalkaloids** are simple amines in which the nitrogen atom is not part of a heterocyclic ring
 - Eg. Ephedrine, taxol and colchicine
- **Pseudoalkaloids** most often have all of the characteristics of the true alkaloids, but they are not derived from amino acids.
 - Eg. Caffeine

True alkaloid need to fulfil the following five points.

- (1) with nitrogen atoms bound in heterocyclic fashion,
- (2) with greater or lesser degree of basic character,
- (3) marked physiological effects,
- (4) complicated molecular structure,
- (5) which are found in plants, and, with a few exceptions, are characteristic for particular plant families, genera, or species"

NB: some true alkaloids do not fulfil all five and also can be from an animal source

Nomenclature

- names of alkaloids should end with the suffix “-ine”
- The names of alkaloids are obtained by various ways:
 - From the specific name of the plant yielding them (**cocaine from *Erythroxylum coca*, belladonnine from *A. belladonna***)
 - From the common name of the drug yielding them (**ergotamine from Ergot**)
 - From the physiologic activity (**emetine (from Greek emetikos, “an emetic”), narcotine, morphine**)
 - Occasionally from the discoverer (**pelletierine after Pelletier**)
 - From the physical property (**hygrine- hygro = moist**)

Distribution

- Alkaloids may be of plant, animal, insect or microbial origin.
- Alkaloids are distributed in 7231 species of higher plants in 1730 genera (~ 14.2 %) within 186 plant families.
 - Monocotyledons: Amaryllidaceae, Liliaceae and Poaceae
 - Dicoltyledon: Annonaceae, Berberidaceae, Papaveraceae, Rubiaceae

Occurrence

- Alkaloids usually occur in plants, in the form of salts of acetic, malic, oxalic, tartaric, succinic, tannic or other plant acids
 - e.g. aconitine occurs in combination with aconitic acid, the opium alkaloids with meconic acid
- Salts with inorganic acids may also be present
 - e.g. morphine is partly present as sulfate in opium.
- A few alkaloids occur in glycosidal combination with sugars, forming the so called **gluco-alkaloids**
 - The toxic **solanine**, found in sprouts of potato tubers

- Alkaloids may occur in every part of a plant, but usually one or more organs have higher content than others.
- Barks, leaves and fruits are often rich in alkaloids, which may occur in particular tissues or cells.
 - Opium alkaloids are encountered specifically in the latex vessels of the opium poppy
 - Tropane alkaloids of *Datura* species are contained in the leaf petiole and in the tissue adjacent to the midrib leaf.
- Alkaloids can be produced at one part of the plant and stored in another
 - Eg: **nicotine** is formed in the root of the tobacco plant and transferred to the leaves.

Properties

- majority of alkaloids are basic compounds and resemble ammonia in chemical properties
 - However, the basicity of alkaloids varies due to availability of the lone pair of electrons on the nitrogen atom
- In the plant they may exist in the free state, as salts or as N-oxides.
- free form are usually soluble only in an organic solvent
 - Exceptions caffeine, colchicine are soluble in water
- Most alkaloids are well-defined crystalline substances which unite with acids to form salts.

Detection

Test	Reagent Composition	Positive Colour Change
Dragendorffs Reagent	Potassium bismuth iodide	Reddish-brown
Mayer's Reagent	Potassio mercuric iodide	White or pale yellow ppt
Hager's Reagent	Picric acid	Yellow
Wagner's Reagent	Solution of iodine in potassium iodide	Yellow or brown ppt.
Murexide Test (for Caffeine and Other Purine Derived Alkaloids)	Potassium chlorate + drops of HCl. Expose the resultant to NH_3	Purple colouration

Extraction

- The general procedures for the isolation of alkaloids are largely based on:
 - The **alkaline nature** of most alkaloids,
 - The subsequent **ability to form salts** with acids and
 - On the **relative solubilities** of the alkaloid bases and their salts **in water** and **in various organic solvents**.
- Defatting the plant material with petroleum ether or hexane may be necessary to remove lipophilic materials (fats, waxes, terpenes, pigments)

- Several extraction methods are possible:

A:Extraction aqueous acid:

1. Adding 2 % Sulphuric acid extracts the alkaloids as their salts form.
2. Salts being water soluble these transfer to the aqueous phase.
3. The acid extract is made alkaline with ammonia solution, shaken gently with an organic solvent in a separating funnel.
4. The alkaloids, as free bases, partition in favor of the organic layer, leaving behind the unwanted non-basic substances.

B. 70–80 % Ethanol or methanol:

1. After extraction the solution can be concentrated to low volume to remove the extraction solvent
2. Acidify with dilute sulphuric acid to form salts of alkaloids
3. Partitioned first with organic solvent to remove unwanted acidic substances (non alkaloid substances)
4. Then basified and re-extracted with fresh organic solvents.
5. This involves an extra purifying step and smaller volume of solvents.

C. Extraction with alkali treatment:

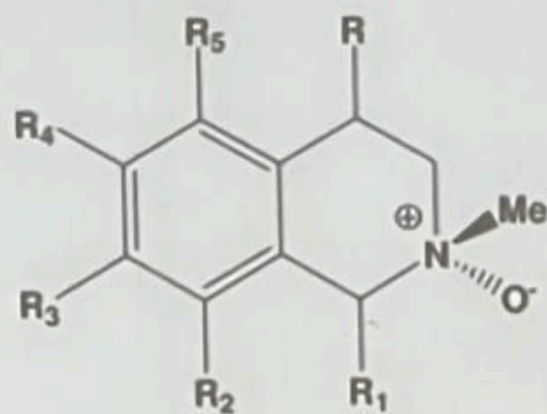
1. The plant material is moistened with aqueous ammonia or calcium hydroxide which converts the alkaloids to the free bases.
2. At the same time, the lime converts organic acids to salts of tannins and other unwanted constituents.
3. The alkaloids are then extracted from the mixture with organic solvent.
4. The extract is purified by shaking with dilute acid, into which the alkaloids pass in the form of salts.
5. Further purification can be achieved by addition of excess alkali to the aqueous solution and renewed extraction with organic solvent

- **Exceptions in extraction of alkaloids**

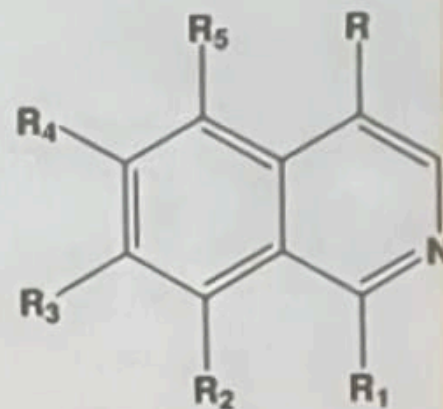
- Alkaloids containing a **quaternary nitrogen** can not usually be extracted with organic solvents from aqueous solution even at alkaline pH.
- They may be precipitated with **picric acid or Mayer's reagent** and converted into soluble bromides or chloride **using ion-exchange chromatography.**
- Besides alkaloids with quaternary nitrogen, **phenolic alkaloids** and **alkaloids N-oxides** cannot be isolated by the standard procedure.

- At high pH phenolic alkaloids are present as **phenolates(anions)** which cannot be extracted with organic solvents.
- Following their liberation from salts by addition of strong base, the pH of the extract must therefore be lowered to a value at which the phenolates **are reduced**, before extraction with an organic solvent can be undertaken.

- Alkaloid N-oxides are more polar than the corresponding alkaloids and are therefore more difficult to extract with an organic solvent.
- They can be extracted with ethyl acetate or chloroform but not with less polar solvents like diethyl ether.



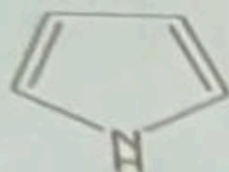
Isoquinoline N-oxide alkaloids



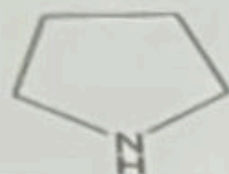
Isoquinoline alkaloids

Structural classification of alkaloids

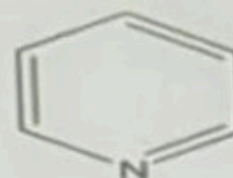
- Classification can be based on:
 - a. Chemical classification
 - b. Pharmacological classification
 - c. Taxonomic classification
 - d. Biosynthetic classification



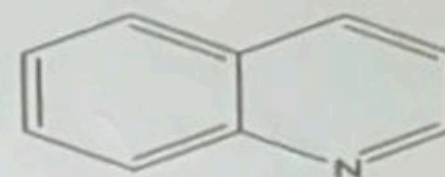
Pyrrole



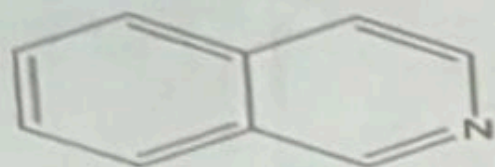
Pyrrolidine



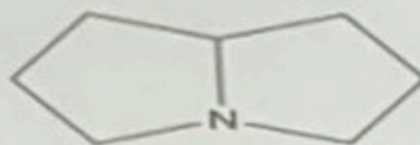
Pyridine



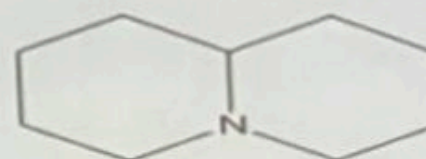
Quinolone



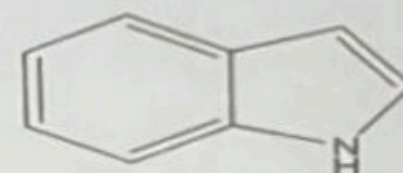
Isoquinoline



Pyrrolizidine



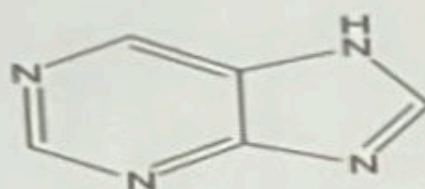
Quinolizidine



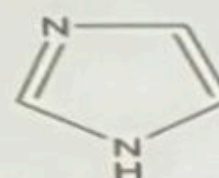
Indole



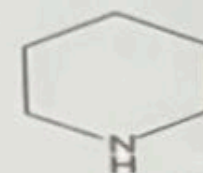
Indolizidine



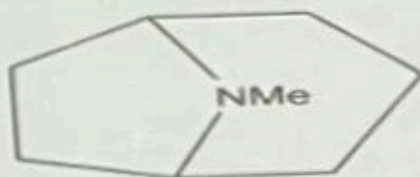
Purine



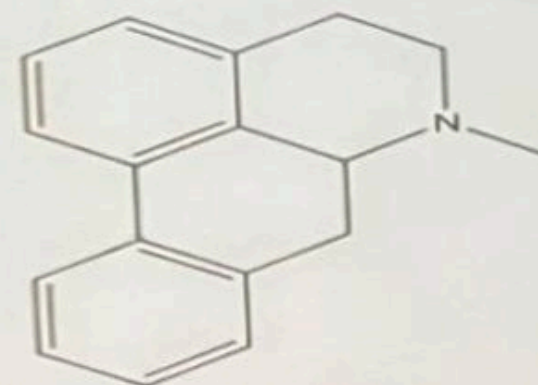
Imidazole



Piperidine



Tropane



Aporphine

- **Chemical classification examples:**

- i. Pyrrolidine alkaloids e.g., **Hygrine**;

- ii. Piperidine alkaloids e.g., Lobeline;

- iii. Pyrrolizidine alkaloids e.g.,

Senecionine;

- iv. Tropane alkaloids e.g., **Atropine**;

- v. Quinoline alkaloids e.g., Quinine;

- vi. Isoquinoline alkaloids e.g., **Morphine**

- vii. Aporphine alkaloids e.g., Boldine;

- viii. Indole alkaloids e.g., **Ergometrine**;

- ix. Imidazole alkaloids e.g., Pilocarpine;

- x. Diazocin alkaloids e.g., Lupanine;

- xi. Purine alkaloids e.g., **Caffeine**;

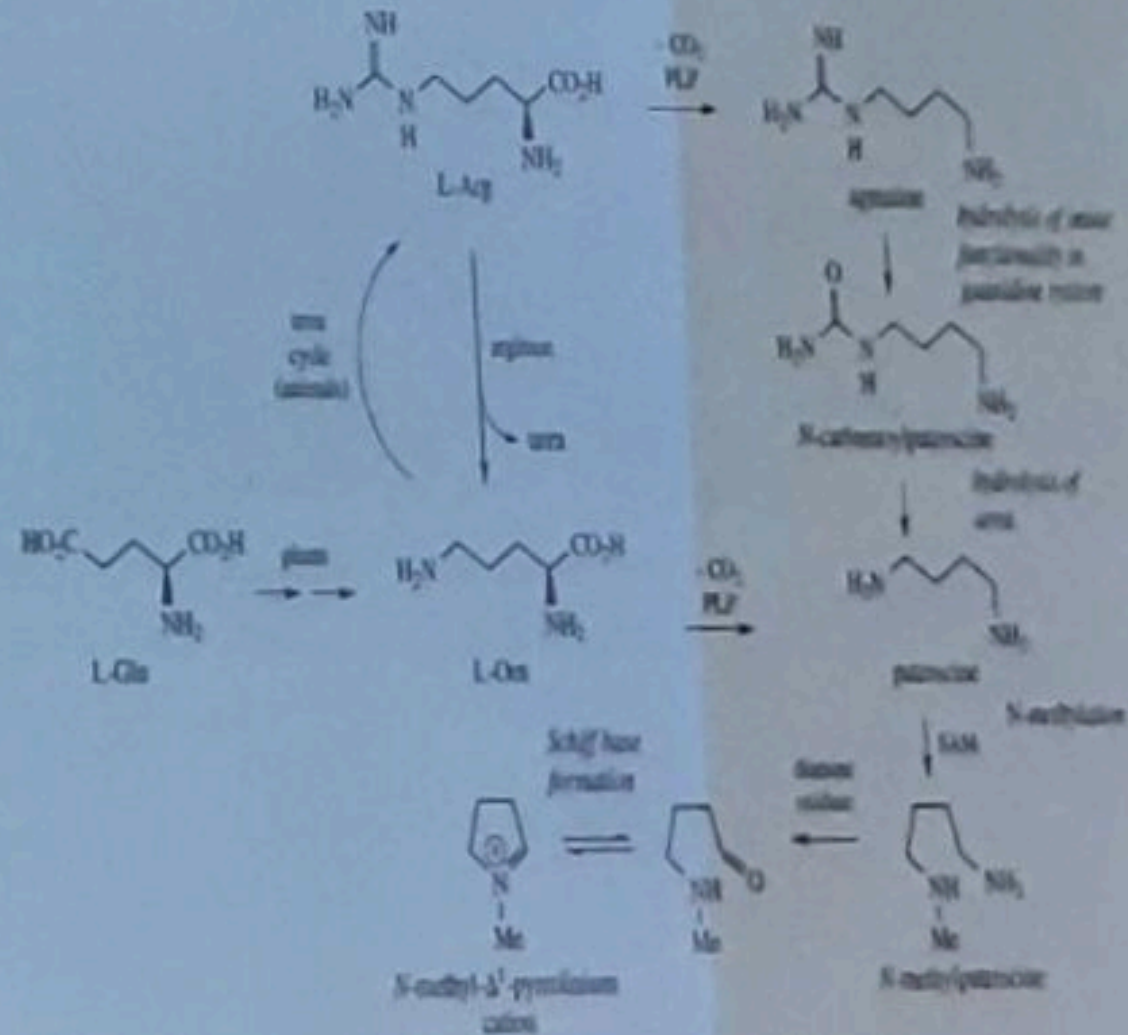
- xii. Steroidal alkaloids e.g., Solanidine;

- xiii. Amino alkaloids e.g., **Ephedrine**;

- xiv. Diterpene alkaloids e.g., Aconitine.

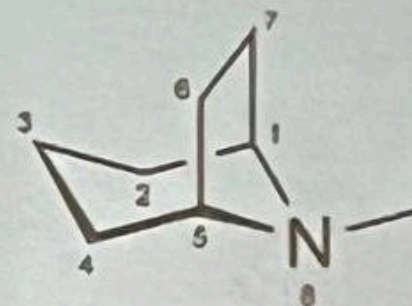
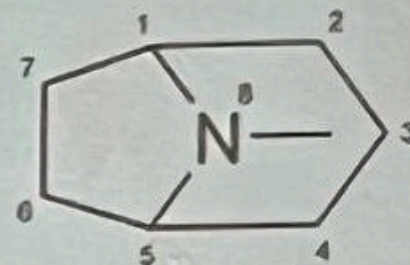
1. Ornithine derived alkaloids

- L-Ornithine is a non-protein amino acid forming part of the urea cycle in animals
- L-Ornithine is an important precursor of **pyrrolidine**, **tropane** and **pyrrolizidine** alkaloids



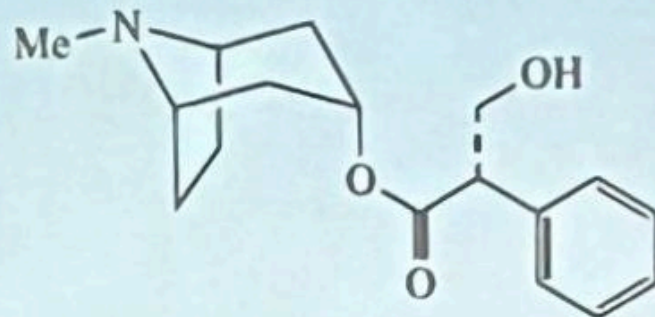
A. Tropane alkaloids

- structurally well-defined group of natural products, and have **mydriatic** and **anesthetic** properties.
- mainly occur in the family Solanaceae from the species of *Datura*, *Atropa*, *Hyoscyamus*, *Scopelia* and *Duboisia*.
- common structural element is the **azabicyclo[3.2.1]octane** system
- over 150 tropane alkaloids have been isolated.
 - Cocaine (*Erythroxylon coca*, coca plant),
 - Hyoscyamine (*Hyoscyamus niger*, henbane),
 - Atropine (*Atropa belladonna*, deadly nightshade),
 - Scopolamine (*Scopola carniolica*)

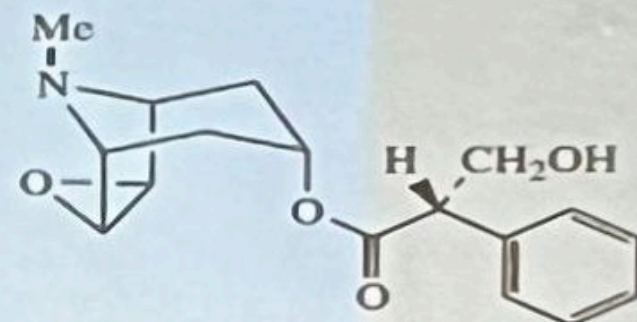


I. Solanaceae alkaloids: Hyoscyamine, scopolamine (hyoscine), and atropine

- (–)-Hyoscyamine and (–)-scopolamine are esters of the amino alcohols tropine and scopine respectively, with (–)-(S)-tropic acid, which is derived from phenylalanine
- The two alkaloids occur exclusively in the Solanaceae family.
- history as psychoactive drugs, poisons, aphrodisiacs, and for the preparation of analgesic and sleeping potions



(–)-hyoscyamine



(–)-hyoscine
(scopolamine)

- Famous examples include the deadly

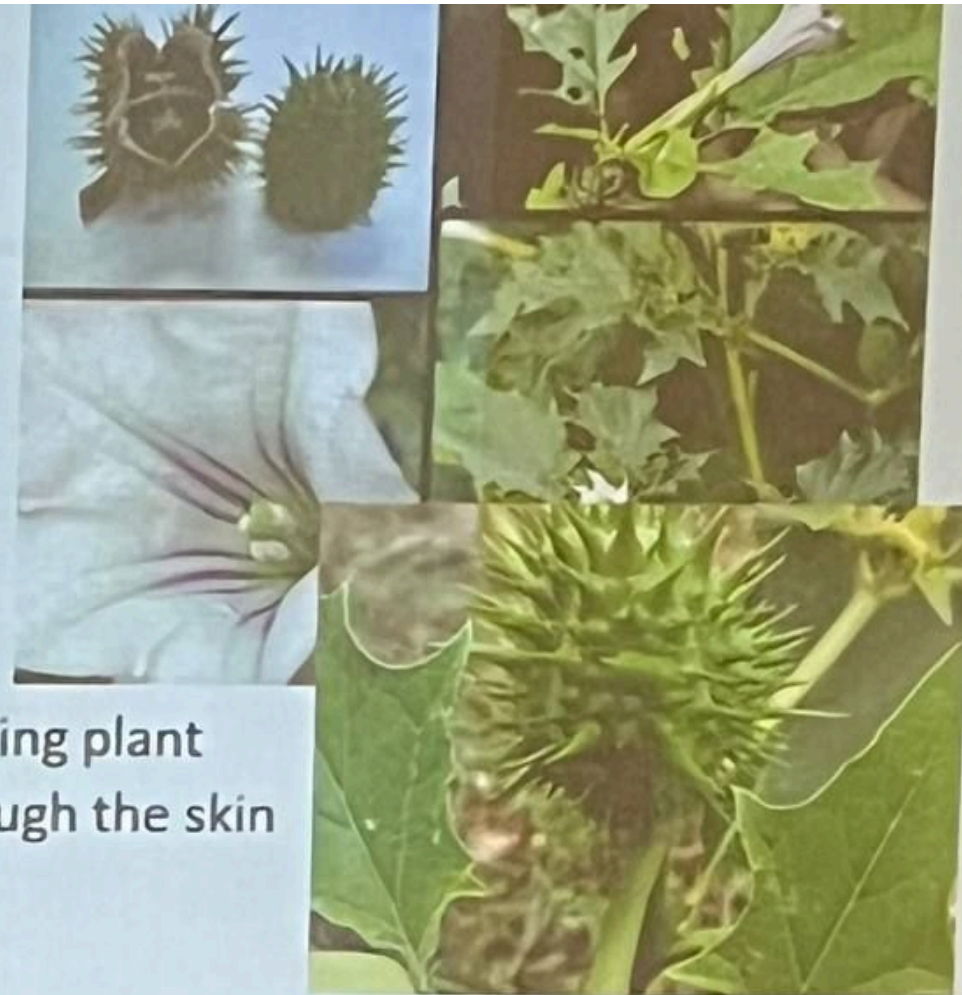
- nightshade (*Atropa belladonna*)
- thorn-apple (*Datura stramonium*)
- henbane (*Hyoscyamus niger*)
- mandrake (*Mandragora officinarum*)
- **All plants are toxic**

- *Atropa belladonna*:

- has long been used as poison and beautifying plant
- Very toxic and can be easily absorbed through the skin
- Common alkaloid is (-)-hyoscyamine

- *Datura stramonium*

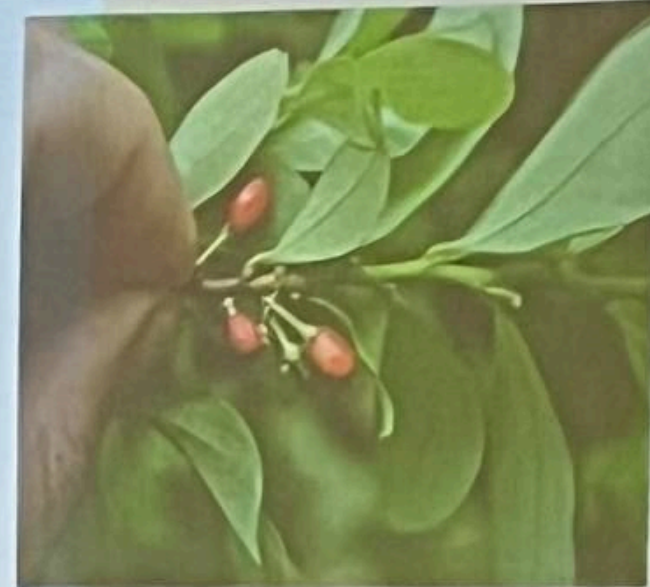
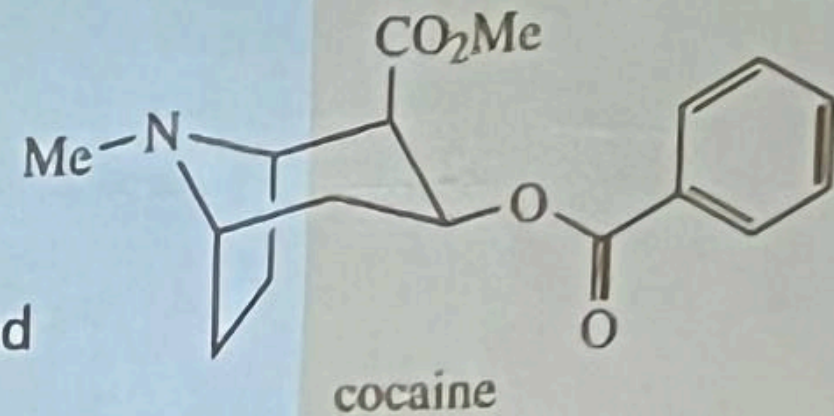
- *In the Middle Ages, stramonium was employed to drug victims prior to robbing them*



- Pharmacological use of hyoscyamine, scopolamine (hyoscine), and atropine:
 - have **anticholinergic effect**: they compete with acetylcholine for the muscarinic site of the parasympathetic nervous system
 - **Antispasmodic action** on the gastrointestinal tract,
 - **Antisecretory effect** controlling salivary secretions during surgical operations, and
 - **As mydriatics** to dilate the pupil of the eye.
 - Unlike hyoscyamine, **scopolamine** is central nervous system depressant and is used prophylactically against **motion sickness**.

II. Coca alkaloids: Cocaine

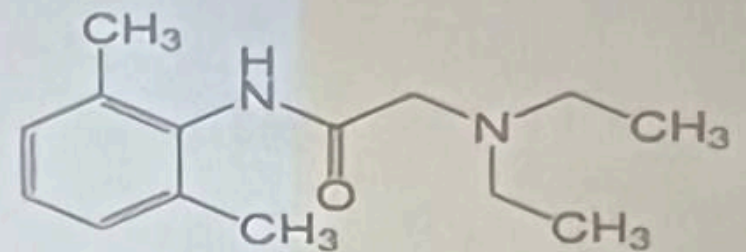
- It is the main component of coca leaves from the species of Erythroxylaceae (*Erythroxylum coca* and *Erythroxylum novogranatense*).
- Cocaine is the benzoic acid ester of the tropane base methylecgonine.
- Cocaine is a **highly addictive central stimulant** that inhibits the re-uptake of the neurotransmitters **dopamine and norepinephrine** at synapses and inhibits monoamine oxidase
- Leading to euphoria, hyperactivity, suppression of hunger and fatigue
- blocks **Na⁺ channels** leading to local anesthesia



- Cocaine derivatives:

- Synthetic drugs developed from the cocaine structure have been introduced to provide safer, less toxic local anaesthetics.
- **Tetracaine** (amethocaine), **oxybuprocaine**, and **proxymetacaine** are valuable local anaesthetics employed principally in ophthalmic work
- **Lidocaine** (lignocaine) is an example of an amino amide analogue and the most widely used local anaesthetic,
 - for treatment of **ventricular arrhythmias** especially after myocardial infarction

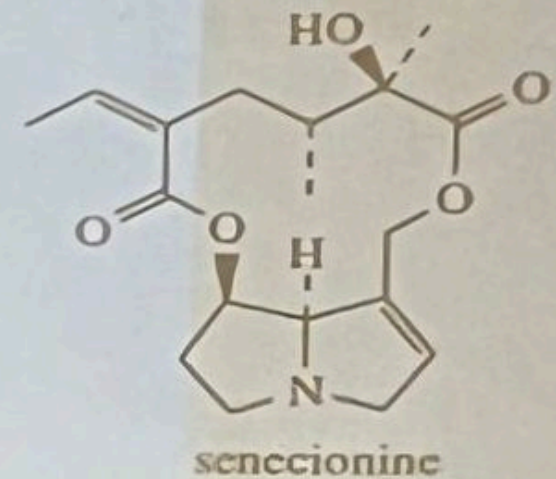
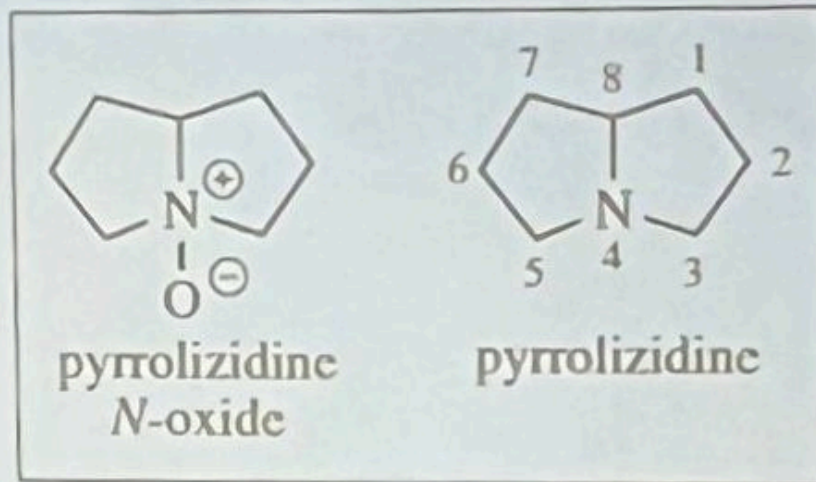
Lidocaine



III. Pyrrolizidine alkaloids

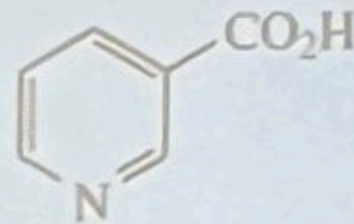
- The backbone of pyrrolizidine alkaloids **Hydroxymethylpyrrolizidine**
- The characteristics of these alkaloids are as follows:
 - accumulated in plants as polar, salt-like N-oxides, facilitating their transport and, maintaining them in a **non-toxic form**;
 - they are poisons;
 - some of them have a bioimpact.
- Many pyrrolizidine alkaloids are **hepatotoxic, mutagenic, and carcinogenic**.

- The structural features responsible for the genotoxicity are:
 - a double bond in the necine base between C-1 and C-2,
 - presence of hydroxyl groups at C-7 and C-8
 - esterification of at least one of these hydroxy groups with a branched carbon chain.

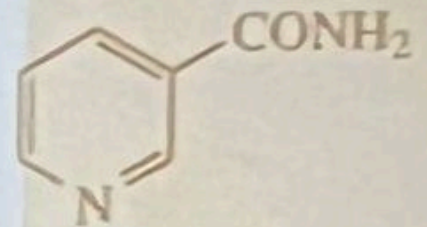


2. Nicotinic acid derived alkaloids: Pyridine alkaloids

- These Alkaloids derived from nicotinic acid (Vitamin B3, niacin) contain a pyridine nucleus



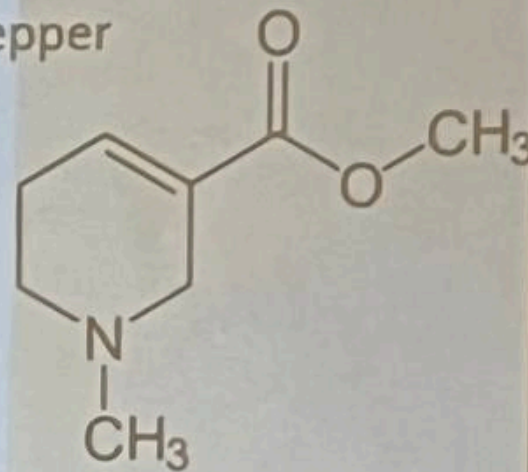
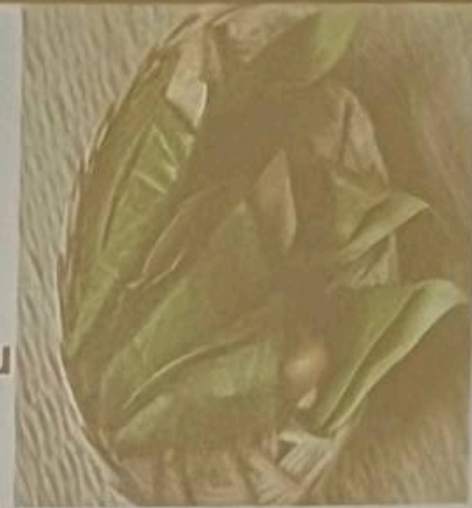
nicotinic acid
(niacin / vitamin B₃)



nicotinamide

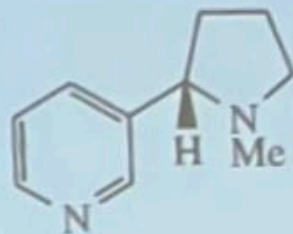
A. Areca alkaloids

- Areca nuts (betel-nuts) are the seeds of the palm tree *Areca catechu* (Palmae/Arecaceae).
- These nuts are mixed with lime, wrapped in leaves of the betel pepper (Piper betle) and then chewed for their stimulant effect:
 - feeling of well-being and mild intoxication
- The major stimulant alkaloid is **arecoline** which is an agonist for muscarinic acetylcholine receptors
- Arecoline has been employed in veterinary practice as a vermicide to eradicate worms.

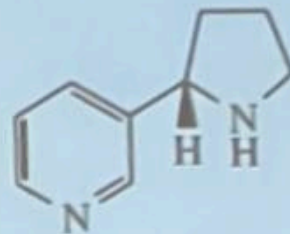


B. Tobacco alkaloids

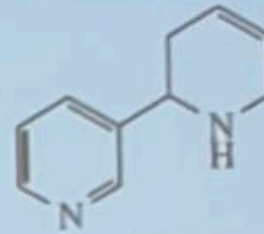
- Tobacco is the cured leaf of tobacco plant *Nicotiana tabacum*
- The principal alkaloids have a pyridine moiety associated with either a **pyrrolidine ring** (ornithine-derived) eg. **nicotine** or a **piperidine ring** (lysine-derived) eg. **anabasine**
- Occur as salts of malic and citric acids.



nicotine



normicotine



anatabine

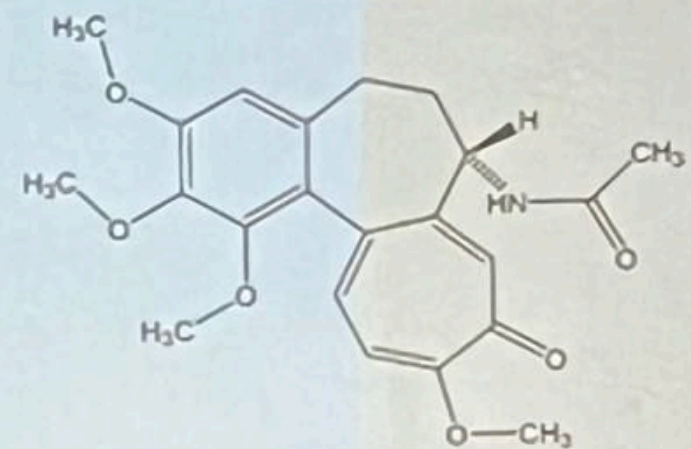
- Nicotine in small doses can act as a **respiratory stimulant**, in larger doses it causes respiratory depression and ultimately death via **respiratory paralysis**.
- Nicotine is toxic to man due to its effect on the nervous system, interacting with the **nicotinic acetylcholine receptors**
- **smoking contributes to, cancer, atherosclerosis, chronic bronchitis, and emphysema**



3. Tyrosine derived alkaloids

A. Colchicine

- Colchicum seed and corm are obtained from *Colchicum autumnale*
- The principal alkaloid is **colchicine**
- Extracts of the plant, and later colchicine itself, have been used in the treatment of gout.
- The nitrogen in colchicine is part of an amide function
 - Less basic and doesn't form salts



B. Isoquinoline alkaloids: Papaver alkaloids

- The opium poppy, *Papaver somniferum*, is the source of the crude drug opium, which contains a large number of alkaloids
 - I. phenanthrene derivatives **morphine** and **codeine**,
 - II. bezylisoquinoline derivative **papaverine**,
 - III. phthalidisoquinoline derivative **noscapi** (**narcotine**).

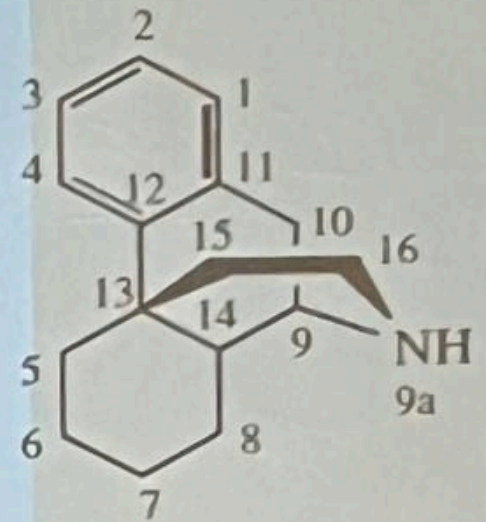


I. Phenanthrene group

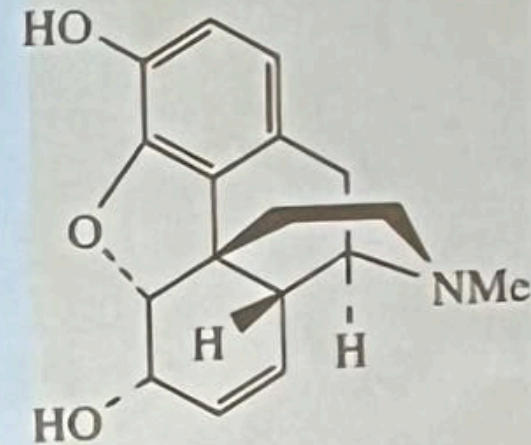
- Morphine, codeine and thebaine are the major alkaloids in this group (morphine type)

Morphine

- has a 4-ring system called **morphinan**.
- It's a powerful analgesic and narcotic, most valuable analgesics for relief of **severe pain**.
- induces a state of **euphoria and mental detachment**
- Has severe withdrawal symptoms thus given for terminal pain



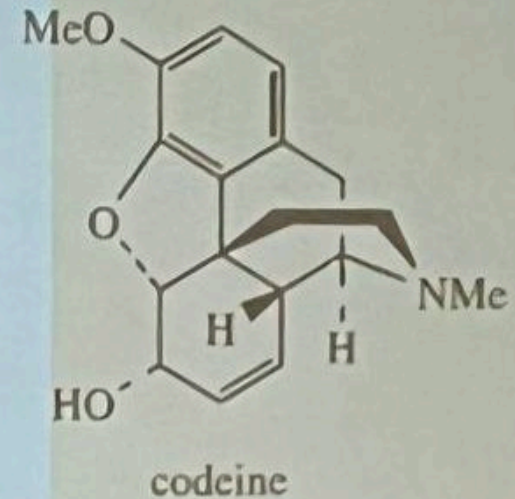
morphinan



morphine

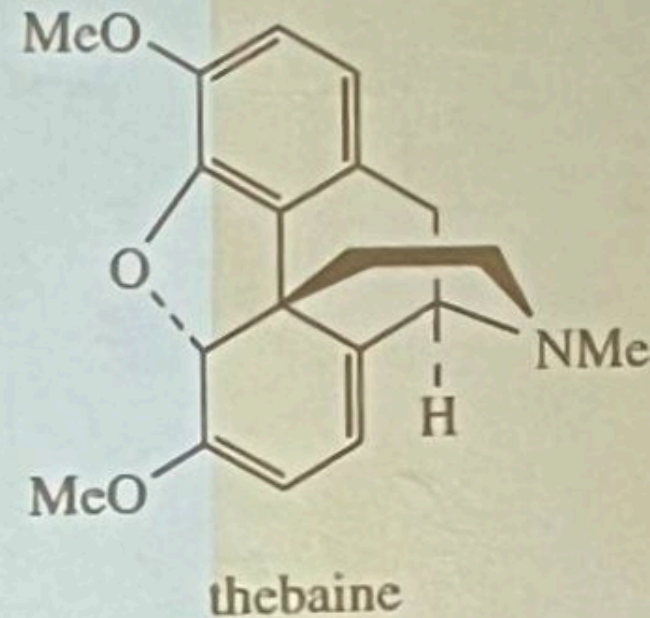
Codeine

- Codeine is the **3-O-methyl ether** of morphine and is the most widely used of the opium alkaloids
- Since its low yield in opium mass produced semisynthetically
- partial **demethylation** in the liver to produce morphine but lacks the **euphoria**
- relatively safe non-addictive medium analgesic with antitussive action



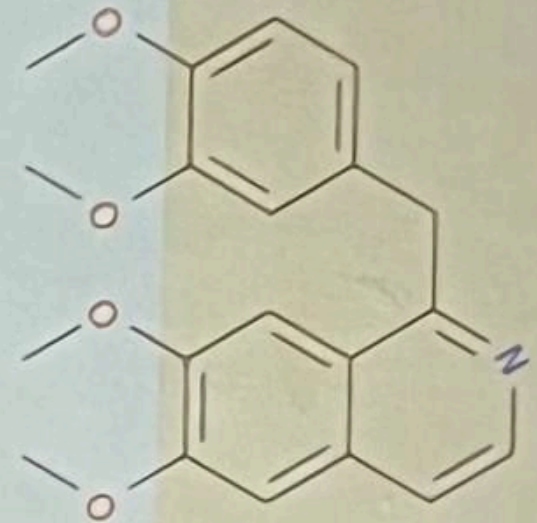
Thebaine

- Thebaine differs structurally from morphine/codeine mainly by its possession of a conjugated diene ring system
- It is almost devoid of analgesic activity and used as **morphine antagonist**
- substrate for the **semi-synthesis** of other drugs such as the **veterinary sedative etorphine**



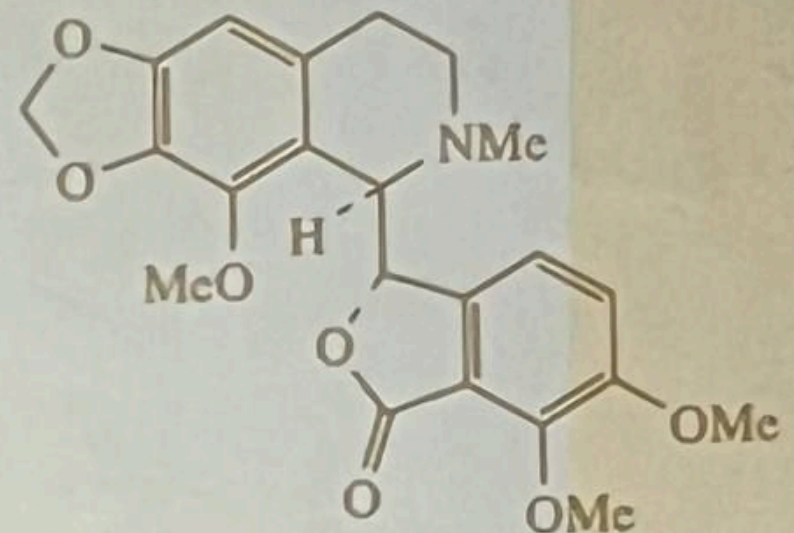
II. Benzyloisoquinoline group: Papaverine

- Papaverine is one of the simplest opium alkaloids
- It is used in medicine as an **antispasmodic**
- It has little or **no analgesic or hypnotic** properties
- relaxes smooth muscle in blood vessels via its Ca^{2+} channel blocking, thus used as **to treat arteial spasm**



III. Phthalide-isoquinoline alkaloids: Noscapine

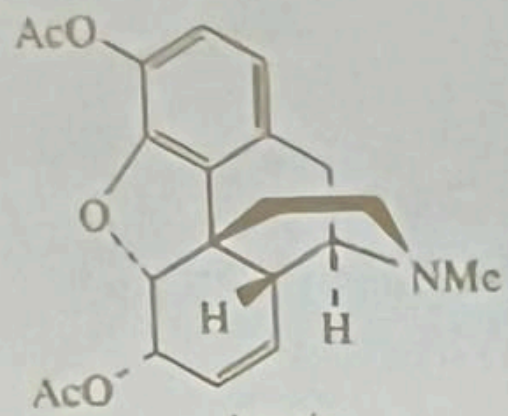
- Noscapine has good antitussive and cough suppressant
- noscapine has **teratogenic properties** (i.e. may deform a fetus) thus band from use



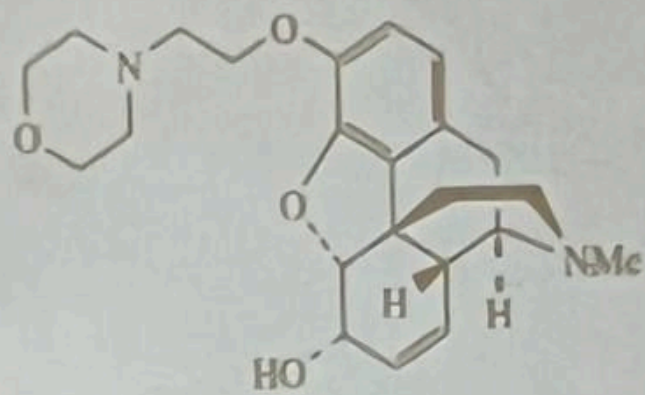
noscapine
(narcotine)

- **Opioids:**

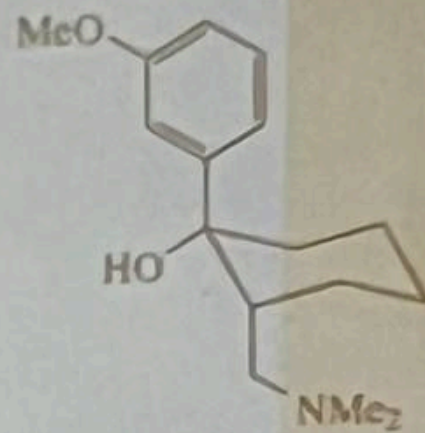
- A vast range of semi-synthetic or totally synthetic morphine-like derivatives
- Same effects as morphine but less addictive
- **Pholcodine** is an effective and reliable antitussive
- **Heroin (Diamorphine)** a Diacetate of morphine although effective cough suppressant highly addictive thus banned from use
- **Pethidine (meperidine)** is one of the most widely used synthetic opiates
 - Pethidine is **less potent than morphine**, but produces prompt, short-acting analgesia
- **Tramadol** is a recent drug claimed to produce analgesia and produces few typical opioid side-effects



heroin
(diacetylmorphine)



pholcodine

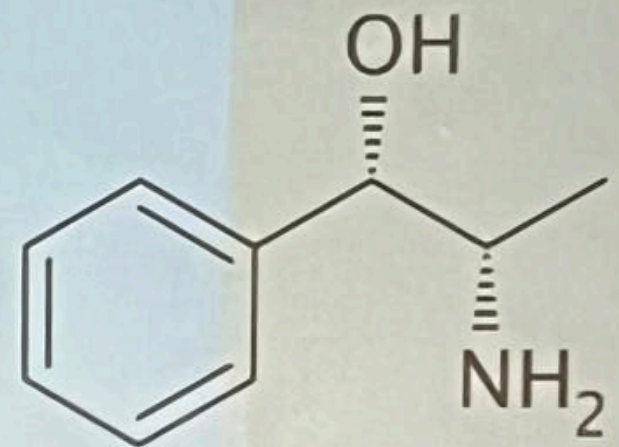
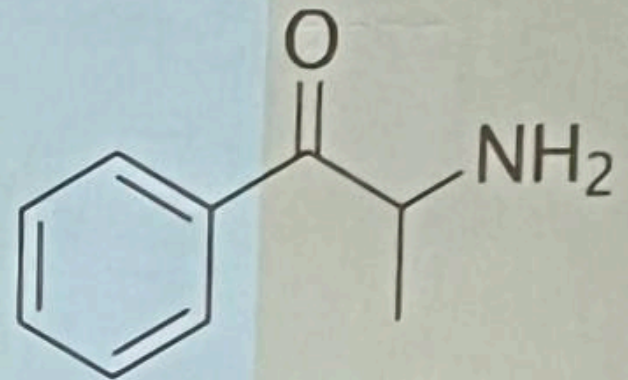


tramadol

4. Phenylalanine derived alkaloids

Alkaloids of "Khat"

- Khat is the fresh leaves of *Catha edulis* a small tree cultivated in Ethiopia, East and South Africa, and in the Yemen.
- Its chewed for a stimulant effect, alleviates hunger and fatigue and general well-being
- Prolonged usage can lead to **hypertension, insomnia, or even mania**
- Young fresh leaves contain 0.1–0.3% **(–)-cathinone** as the principal central nervous system stimulant
- Cathinone is **relatively unstable, decomposing** to **(+)-norpseudoephedrine (cathine)** and **norephedrine** after harvesting or as the leaves are dried



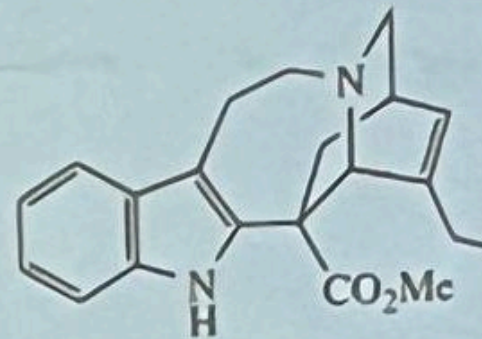
5. Tryptophan derived alkaloids

Periwinkle alkaloids

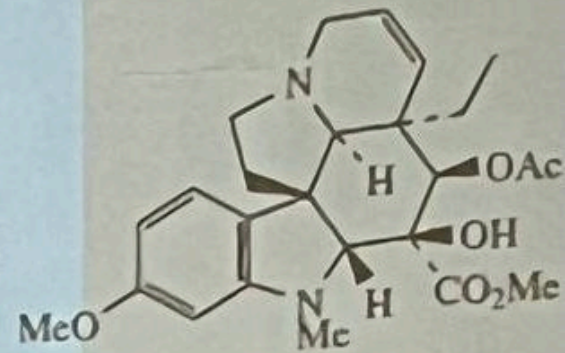
- The Madagascar periwinkle *Catharanthus roseus*
- More than 150 alkaloids have since been characterized in the plant
- principally **terpenoid indole alkaloids**
- **Vinblastine** and **vincristine** are well know anti-tumor drugs
- These compounds all inhibit cell mitosis, acting by binding to the **protein tubulin in the mitotic spindle, preventing polymerization into microtubules**
- The yield was very low and this created a major obstacle for their clinical use



- partial synthesis for the dimeric indole alkaloids was developed starting from the monomers **vindoline** and **catharanthine**

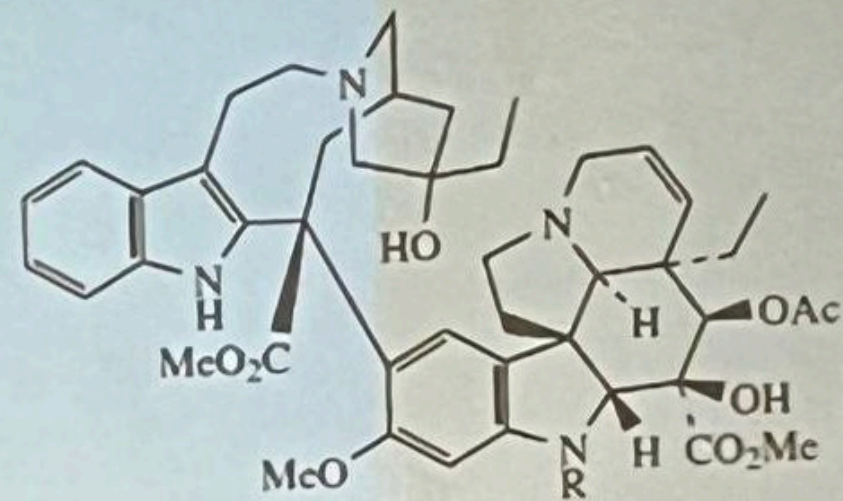


catharanthine



vindoline

- **Vinblastine** is used mainly in the treatment of **Hodgkin's disease**, a cancer affecting the lymph glands, spleen, and liver.
- **Vincristine** has superior antitumour activity compared with vinblastine but is **more neurotoxic**



R = Me, vinblastine
R = CHO, vincristine

Reading assignment

- Quinoline Alkaloids
 - Quinine